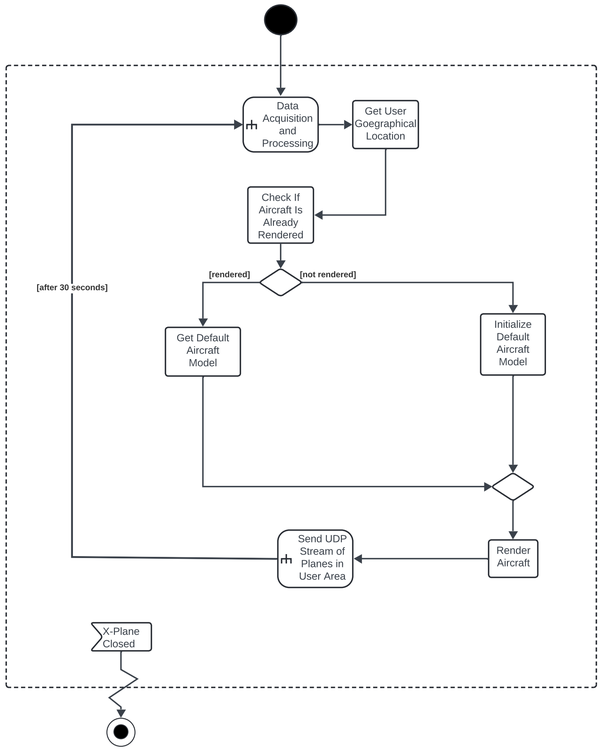
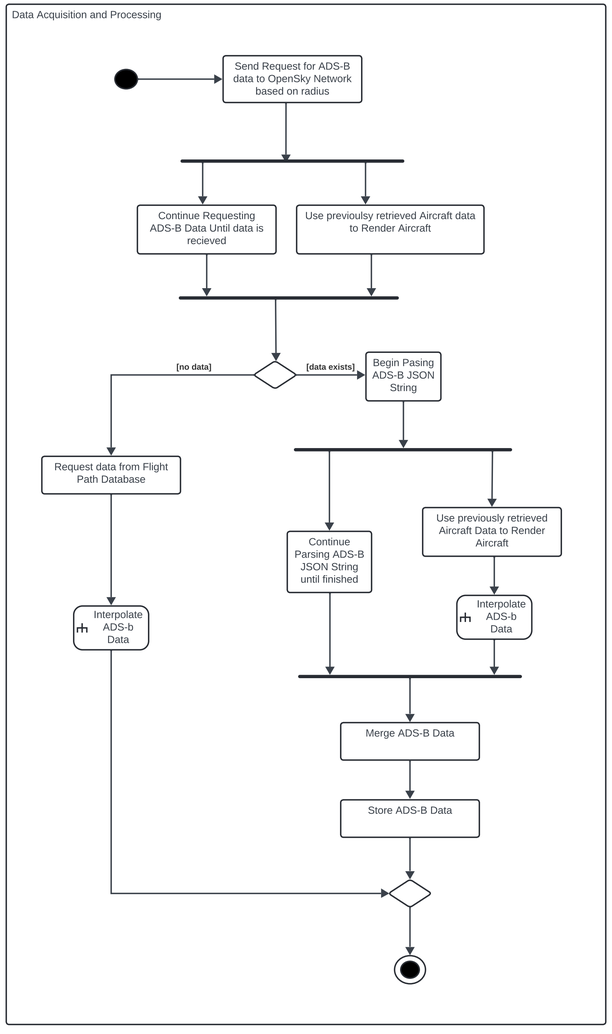
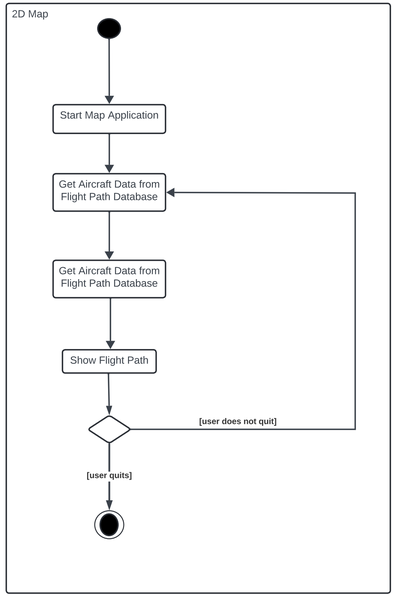
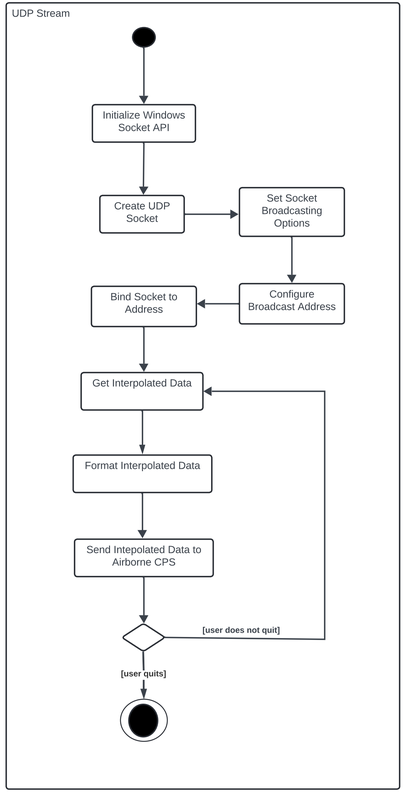

Data Acquisition and Processing
Get User Goegraphical Location
Check If Aircraft Is Already Rendered
[rendered]
[not rendered]
[after 30 seconds]
Get Default Aircraft Model
Initialize Default Aircraft Model
Render Aircraft
Send UDP Stream of Planes in User Area
X-Plane Closed


Data Acquisition and Processing
Send Request for ADS-B data to OpenSky Network based on radius
Continue Requesting ADS-B Data Until data is recieved
Use previoulsy retrieved Aircraft data to Render Aircraft
[no data]
[data exists]
Begin Pasing ADS-B JSON String
Request data from Flight Path Database
Continue Parsing ADS-B JSON String until finished
Use previously retrieved Aircraft Data to Render Aircraft
Interpolate ADS-b Data
Interpolate ADS-b Data
Merge ADS-B Data
Store ADS-B Data


2D Map
Start Map Application
Get Aircraft Data from Flight Path Database
Get Aircraft Data from Flight Path Database
Show Flight Path
[user does not quit]
[user quits]


UDP Stream
Initialize Windows Socket API
Create UDP Socket
Set Socket Broadcasting Options
Bind Socket to Address
Configure Broadcast Address
Get Interpolated Data
Format Interpolated Data
Send Intepolated Data to Airborne CPS
[user does not quit]
[user quits]